

Curriculum Loss Strategy

Epochs 1 – 5 (Warm-up)

switch at epoch 5

Epoch 6 onward (Main training)

Huber Loss ($\delta = 1.0$)

$\frac{1}{2}(y - \hat{y})^2$ if $|y - \hat{y}| < 1$

$|y - \hat{y}| - \frac{1}{2}$ otherwise

*Protects against outlier sensors
during unstable early training*

Masked MAE Loss

$(1/|\Omega|) \sum |y - \hat{y}|, \Omega = \{|y| > 0\}$

*Directly minimises the evaluation metric.
MAPE threshold: $|y| > 10$ vehicles / 5 min
to exclude near-zero night-time counts.*

Optimiser: AdamW | lr = 10^{-3} | weight_decay = 10^{-4} | gradient clip norm = 5.0 | AMP enabled

Scheduler: 5-epoch linear warm-up → Cosine Annealing WarmRestarts ($T_0 = 50$) + ReduceLROnPlateau (patience = 8, factor = 0.5)